Measure sound speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Analyze sound travel measurements and their limitations.

SUCCESS CRITERIA - I can...

1. Define one-way versus round-trip paths.
2. Calculate speed from distance and time.
3. Use repeated measurements and discuss timing uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Three instantaneous displacements are +3 mm, -5 mm and -7 mm. Find their sum and explain why one displacement value does not determine total wave energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: v = d/Δt; v = 2 d/Δt for an echo.
Match the distance to one-way or round-trip travel. Use moderate sound levels and record actual data separately.

Worked reasoning: A sound echo returns 0.4 s after a pulse. At 340 m/s, round-trip distance is 136 m, so the reflector is 68 m away.

Your turn: Draw outgoing and returning paths and mark where the factor of 2 enters.



Marzano check - circle: ① ② ③

03 Read and interpret the model

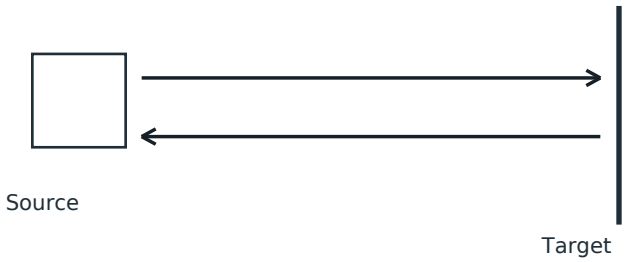
A pulse travels between two sensors separated by 1.70 m. Delay is 0.0050 s. Is this one-way or echo distance? Find speed.



Marzano check - circle: ① ② ③

04 Apply the physics

A microphone detects an echo 0.0200 s after a pulse hits the air. The reflecting wall is 3.40 m away. Find sound speed. Treat distance as exact and timing as 0.0200 ± 0.0005 s; calculate the lowest and highest possible speeds. Label the round trip.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

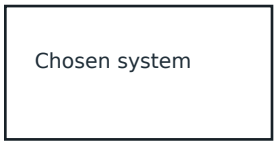
Plan a table for three trials. Include distance, elapsed time, method (one-way or echo), speed and one possible systematic error.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A sphere’s radius becomes 4 times as large at fixed density. Derive the area-to-mass multiplier. Explain why kinetic energy per mass alone does not predict survival during entry.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Two sensors 0.85 m apart record a wave’s travel time as 0.0025 s. Find the wave speed. A timing fault adds 0.001 s to the recorded interval. Calculate the speed this faulty measurement would give, and explain the direction of the error. Label the one-way path.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A microphone pair separated 1.5 m measures travel-time delay 0.0044 s. Find speed. If delay is uncertain by±0.0002 s, find worst-case speed bounds.



Marzano check - circle: 1 2 3